

Structural Re-Parameterization, Spatial Coordinate Encoding, and Cross-Domain Robustness for Real-Time Safety Helmet Detection in Construction Surveillance

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Abstract—Automated Personal Protective Equipment (PPE) compliance monitoring via CCTV is vital for injury prevention on construction sites. However, edge surveillance faces severe operational bottlenecks: distant tiny helmet targets ($< 20 \times 20$ pixels), extreme category skew (1 : 12 helmet-to-person ratio), multi-scale occlusions, and background clutter. We present Rep-YOLO11s, a domain-specialized detector engineered as a principled Hardware-Software Co-Design for real-time edge surveillance. Guided by the GPU Roofline model, our architecture resolves the memory-bandwidth bottlenecks of single-batch inference while targeting four core physical constraints of construction monitoring: (1) Memory Access Cost (MAC) minimization via Structural Re-parameterization (RepConv) multi-branch training blocks folded algebraically into single 3×3 convolutions at inference; (2) vertical spatial inductive priors via Stem CoordConv (+0.06 ms) resolving ground distractor confusion; (3) Bi-Level Routing Attention (BiFormer, $S = 8, k = 4$) pruning 93.75% of irrelevant background tokens in neck stages (+0.24 ms) without quadratic complexity; and (4) decoupled multi-task loss pairing Focal EIoU Loss for boundary quality with Task-Aligned Assignment (TAL) for severe 1:12 class imbalance. On the Safety Helmet Wearing Dataset (SHWD / VOC2028), Rep-YOLO11s achieves 94.83% mAP_{50} and 62.54% mAP_{50-95} under joint two-class evaluation, with 5-fold cross-validation reaching a mean mAP_{50} of 96.64% (standard deviation 0.32%, peak 97.11%). Under the Harmonized Hat-Only Protocol, it achieves 77.90% mAP_{50-95} at 640×640 and 78.93% at 960×960 (+4.80% over baseline). In matched-engine deployment audits, batch-one TensorRT FP16 execution at 640×640 achieves 2.92 ms (342.5 FPS) on NVIDIA Tesla T4, 4.37 ms (228.6 FPS at 28.4 W) on RTX 3050 Laptop GPU (182.4 FPS / 12.16 Frames/J under a 15 W power cap), and real-time execution on an entry-level GeForce MX230 (36.0 ms, 27.8 FPS). Zero-shot evaluations across GDUT-HWD, Hard Hat Workers, and SHD confirm robust cross-domain transferability under harmonized PPE protocols.

Index Terms—Safety Helmet Detection, Bi-Level Routing Attention, Structural Re-parameterization, CoordConv, Focal EIoU Loss, Cross-Domain Generalization, Real-Time Edge AI.

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I. INTRODUCTION

Personal Protective Equipment (PPE) compliance monitoring, particularly for safety helmets, is strictly mandated across industrial construction sites to prevent traumatic head injuries. While automated intelligent video analytics via CCTV streams offer continuous, non-intrusive compliance auditing, deploying computer vision models in real-world construction environments encounters four major operational bottlenecks: (1) **Distant Tiny Targets**: long-range CCTV cameras render helmets as sub- 20×20 pixel targets subject to severe feature attenuation; (2) **Severe Category Skew**: industrial benchmarks like SHWD exhibit acute category skew (9,044 helmet vs. 111,514 person instances, a 1 : 12 ratio), biasing standard losses; (3) **Spatial Distractor Confusion**: round yellow buckets and orange cones trigger false alarms under translation-invariant convolutions lacking vertical context; and (4) **Strict Edge Latency Bounds**: multi-camera streams require real-time throughput (≥ 60 FPS), penalizing complex attention with high Memory Access Cost (MAC).

To overcome these barriers, we formulate **Rep-YOLO11s** as a principled Hardware-Software Co-Design for edge video surveillance guided by the GPU Roofline model. Rather than treating architectural enhancements as disconnected heuristics, our design directly targets the memory-bandwidth bottleneck of single-stream ($N_{\text{batch}} = 1$) edge inference, where operational intensity (18–28 FLOPs/byte) sits an order of magnitude below the compute ceiling. Our scientific contributions are: (1) **Roofline-Guided RepConv Structural Fusion**: Multi-branch training topologies (3×3 , 1×1 , and identity strictly guarded by $\mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}}$) fold algebraically into single 3×3 kernels (switch_to_deploy) for inference, eliminating intermediate DRAM roundtrips and multi-kernel launch overheads. (2) **Stem CoordConv Inductive Priors**: Normalized coordinates $(x, y) \in [-1, 1]$ injected strictly at the initial Stem layer (5 $\rightarrow$ 64) supply vertical positional priors at negligible cost (+0.06 ms), differentiating head-worn helmets from ground distractors without intermediate tensor inflation. (3) **Dynamic Sparse Attention with Native TRT Export**:

Dynamic bi-level routing attention ($S = 8, k = 4$) in neck stages (P_4, P_5) filters out 93.75% of irrelevant background tokens, exported directly to ONNX Opset 17 via native TopK/Gather without custom C++ plugins (+0.24 ms). (4) **Decoupled Multi-Task Imbalance Supervision:** Bounding box regression quality imbalance is resolved via Focal EIoU ($\gamma = 0.5$), which eliminates aspect-ratio gradient freezing, while acute 1 : 12 category imbalance is decoupled and handled by Task-Aligned Assignment (TAL) and aligned classification loss. (5) **Matched-Engine Empirical Audits:** Validated on SHWD (94.83% mAP_{50} ; 5-fold CV 96.64%) and three external domains under harmonized protocols. Benchmarked with strict stream synchronization across Tesla T4, RTX 3050 Laptop GPU, and entry-level MX230 across matched PyTorch and TensorRT FP16 engines.

II. RELATED WORK

A. Real-Time Object Detection Frameworks

Real-time single-stage object detection has evolved through the YOLO series [1]–[3]. YOLOv6 [4] introduced structural re-parameterization via EfficientRep backbones and Rep-PAN. YOLOv8 adopted anchor-free C2f modules, while YOLOv10 [2] explored dual-assignment NMS-free training. Ultralytics YOLO11 [3] refined pyramid alignment via C3k2 and C2PSA blocks. Recently, YOLO26 [5] explored dual-label assignment for NMS-free and DFL-free edge deployment, while Visual State Space Models (e.g., VMamba [6]) adopt linear sequence modeling for vision. However, in dense construction scaffolding with severe occlusion, NMS-free one-to-one matching risks recall degradation on overlapping tiny helmets, while 1D sequence serialization in SSMs sacrifices 2D spatial locality unless augmented with multi-directional scans. Rep-YOLO11s couples Task-Aligned Assignment (TAL) with Bi-Level Routing Attention (BiFormer) to preserve 2D local context while bounding latency.

B. Safety Helmet Detection in Complex Environments

Various architectures have tailored detectors for PPE auditing. EC-YOLOv8 [7] incorporated ECA attention, CARAFE upsampling, and EIoU loss into YOLOv8n (95.7% mAP_{50} on SHWD), but CARAFE reduces speed to 172 FPS on desktop GPUs and remains sensitive to distractors. Wu *et al.* [8] proposed YOLO-CBF combining CoordConv, BiFormer attention, and Focal-EIoU on a small 962-image road dataset without validating against industrial clutter. Zhao *et al.* [9] used DCNv2 and RCF for power-grid inspection, but DCNv2 incurs irregular memory access that penalizes TensorRT. Song *et al.* [10] combined DWR modules, ASPP, and NWD loss, but parallel dilated branches expand memory footprint. Additional variants including DABFNet [11] and MAF-YOLO [12] explore attention and re-parameterization, yet either incur quadratic attention overhead or fail to decouple multi-branch training topologies from edge execution kernels, causing severe Memory Access Cost (MAC).

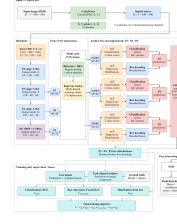


Fig. 1. Overall Architecture of Rep-YOLO11s. The pipeline integrates: (1) Stem CoordConv coordinate concatenation ($[RGB, X, Y] \rightarrow 5 \times 640 \times 640$); (2) Multi-scale backbone feature extraction (P1 Stem, P2–P4 C3k2, P5 SPPF+C2PSA); (3) PAN neck abstraction with Bi-Level Routing Attention (BRA) and RepConv; (4) Anchor-free decoupled detection heads (P3, P4, P5) predicting classification logits ($cv3$) and distribution regression ($cv2$); and (5) Multi-task loss supervision with Task-Aligned Assigner (TAL), BCE, Focal-EIoU, and DFL.

C. Structural Re-parameterization and Spatial Encoding

Structural re-parameterization, pioneered by RepVGG [13] and adopted in YOLOv6 [4], decouples training representation from inference execution. Parallel branches ($3 \times 3, 1 \times 1$, identity) provide gradient diversity during backpropagation and fold algebraically into single 3×3 convolutions at deployment, eliminating residual branching and reducing Memory Access Cost (MAC). CoordConv [14] breaks translation equivariance by concatenating coordinate channels, providing vertical spatial priors critical for distinguishing helmets on human heads from ground-level construction materials.

III. PROPOSED METHODOLOGY

A. Overall System Architecture

The architecture of Rep-YOLO11s is illustrated in Fig. 1. The framework adopts an end-to-end single-stage design developed from Ultralytics YOLO11 [3], comprising: (1) a Structural Re-parameterized Backbone enhanced with Stem CoordConv; (2) a Path Aggregation Network (PAN) Neck [16], [17] integrated with BiFormer Bi-Level Routing Attention; and (3) an Anchor-Free Decoupled Detection Head trained with Focal EIoU Bounding Box Loss and Task-Aligned Assignment.

B. Structural Re-parameterization (RepConv) Mechanics

To achieve high non-linear feature representation during training without incurring latency penalties during edge deployment, RepConv blocks are embedded within critical neck aggregation pathways.

During the training phase (Fig. 2), a RepConv layer with input channels C_{in} , output channels C_{out} , and stride s is formulated across three parallel branches:

$$y = \text{BN}_{3 \times 3}(W^{3 \times 3} * x) + \text{BN}_{1 \times 1}(W^{1 \times 1} * x) + \mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}} \cdot \text{BN}_{id}(x), \quad (1)$$

where $W^{3 \times 3} \in \mathbb{R}^{C_{out} \times C_{in} \times 3 \times 3}$ and $W^{1 \times 1} \in \mathbb{R}^{C_{out} \times C_{in} \times 1 \times 1}$ represent convolution kernels, and BN denotes batch normalization parameters $\{\mu, \sigma^2, \gamma, \beta\}$. Crucially, the identity mapping branch $\text{BN}_{id}(x)$ exists if and only if $C_{in} = C_{out}$ and stride $s = 1$. When $C_{in} \neq C_{out}$ or $s > 1$, no identity branch is instantiated ($\mathbb{I} = 0$), strictly ensuring dimensional

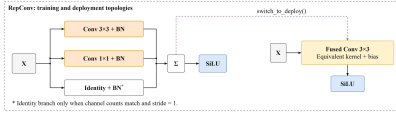


Fig. 2. RepConv Structural Re-Parameterization Lifecycle [13]. Multi-branch training topology (left) is algebraically folded into a single 3×3 kernel (right) for deployment.

and stride consistency without requiring additional projection convolutions.

During deployment conversion (`switch_to_deploy`), each Conv-BN sequence is first collapsed into an equivalent kernel W' and bias vector b' :

$$W'_{i,:,:,} = \frac{\gamma_i}{\sqrt{\sigma_i^2 + \epsilon}} W_{i,:,:,}, \quad b'_i = \beta_i - \frac{\gamma_i \mu_i}{\sqrt{\sigma_i^2 + \epsilon}}. \quad (2)$$

Next, $W'_{1 \times 1}$ is zero-padded into a 3×3 kernel $W'_{1 \times 1 \rightarrow 3 \times 3}$ at center position (2, 2). When $C_{in} = C_{out}$ and $s = 1$, the identity branch transforms into an equivalent 3×3 Dirac delta kernel W'_{id} where $W'_{id,c,c,2,2} = 1$ and 0 elsewhere, folded with BN_{id} ; when $C_{in} \neq C_{out}$ or $s > 1$, $W'_{id} = 0$ and $b'_{id} = 0$. Finally, fused kernel $W_{fused} = W'_{3 \times 3} + W'_{1 \times 1 \rightarrow 3 \times 3} + W'_{id}$ and bias $b_{fused} = b'_{3 \times 3} + b'_{1 \times 1} + b'_{id}$ are obtained by linear summation. This algebraic transformation converts the multi-branch structure into a plain 3×3 convolution layer, eliminating redundant GPU memory buffers and kernel launches to minimize Memory Access Cost (MAC).

C. Spatial Coordinate Convolution (CoordConv)

Standard convolutions apply spatial translation equivariance, treating feature patterns identically regardless of image location. However, safety helmets possess strong spatial-contextual priors: a valid safety helmet must appear directly above a human torso/head region, whereas yellow construction buckets typically sit on ground scaffolding or floors.

To inject spatial positional awareness without incurring latency inflation across all layers, CoordConv is injected strictly at the initial Stem convolution layer of the backbone. We augment the raw RGB input image $X \in \mathbb{R}^{3 \times H \times W}$ with two normalized coordinate channels $C_x(i, j) = \frac{2j}{W-1} - 1$ and $C_y(i, j) = \frac{2i}{H-1} - 1$. The concatenated 5-channel tensor $X_{Coord} = [X; C_x; C_y] \in \mathbb{R}^{5 \times H \times W}$ is passed to the Stem Conv layer ($c_1 = 5 \rightarrow c_2 = 64, k = 3, s = 2$). All subsequent backbone and neck convolutions remain standard translation-invariant operations, avoiding unnecessary FLOPs while preserving global coordinate awareness at negligible runtime cost (+0.06 ms).

D. Bi-Level Routing Attention (BiFormer)

Distant helmets in crowded scenes contain fine-grained details easily masked by background clutter. Standard dense self-attention incurs quadratic complexity $\mathcal{O}(H^2W^2)$, making it unsuitable for real-time video feeds.

We employ Bi-Level Routing Attention (BRA) [15] embedded within high-level neck stages ($P_4/16$ and $P_5/32$) as shown in Fig. 3. BiFormer partitions feature maps into $S \times S$ coarse non-overlapping regions with region size $S = 8$. For region-level queries Q^r and keys K^r (obtained via average pooling

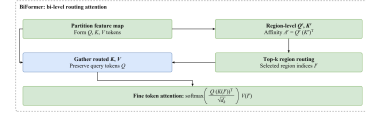


Fig. 3. Bi-Level Routing Attention (BRA) Architecture [15]. Coarse region partitioning ($S = 8, k = 4$) filters out 93.75% irrelevant tokens before computing fine-grained token attention.

per region), coarse region affinities $A^r = Q^r(K^r)^T/\sqrt{C}$ determine routing indices $I^r = \text{TopK}(A^r, k = 4)$ keeping only the top- k ($k = 4$) most relevant regions per query region. Fine-grained token-level attention is computed across 4 attention heads (head dimension $d_k = C/4$, expansion ratio $e = 2$) exclusively across gathered key-value tokens: $\text{Attention}(Q, K, V) = \text{Softmax}(Q(K^{(I^r)})^T/\sqrt{d_k})V^{(I^r)}$. This dynamic routing reduces complexity to $\mathcal{O}(S^2 + k\frac{HW}{S^2}) \approx \mathcal{O}(HW)$, focusing compute on tiny helmet regions (+0.24 ms TRT overhead). For deployment, our BiFormer maps strictly onto native ONNX Opset 17 operators (TopK, GatherElements, Gemm, Softmax), enabling direct TensorRT FP16 engine compilation via standard `trtexec` without custom runtime extensions.

E. Bounding Box Regression and Multi-Task Loss Supervision

Bounding box regression in standard YOLO uses Complete IoU (CIoU) loss. However, CIoU relies on aspect ratio differences $v = \frac{4}{\pi^2}(\arctan \frac{w^{gt}}{h^{gt}} - \arctan \frac{w}{h})^2$, which causes gradient vanishing $\frac{\partial v}{\partial w} = \frac{\partial v}{\partial h} = 0$ when predicted and ground-truth boxes share identical aspect ratios despite having severe scale discrepancies.

We implement Focal EIou Loss [19], which explicitly decomposes bounding box discrepancy into overlap L_{IoU} , distance L_{dis} , and side length L_{asp} :

$$L_{EIou} = 1 - \text{IoU} + \frac{\rho^2(\mathbf{b}, \mathbf{b}^{gt})}{c^2} + \frac{(w - w^{gt})^2}{C_w^2} + \frac{(h - h^{gt})^2}{C_h^2}, \quad (3)$$

where $\mathbf{b}, \mathbf{b}^{gt}$ represent central points, $\rho(\cdot)$ is Euclidean distance, c is diagonal length of the smallest enclosing box, and C_w, C_h are enclosing box width and height. Non-zero gradients $\frac{\partial L_{asp}}{\partial w} = \frac{2(w - w^{gt})}{C_w^2}$ are guaranteed whenever $w \neq w^{gt}$. To prioritize hard-sample localization under heavy occlusion, a focal weighting factor is applied: $L_{\text{Focal-EIoU}} = \text{IoU}^\gamma \cdot L_{EIou}$ with $\gamma = 0.5$.

Crucially, while Focal EIou governs geometric boundary regression, the acute 1 : 12 category imbalance between helmet (9,044) and person (111,514) instances is decoupled and resolved by the Task-Aligned Assigner (TAL) [20] and aligned classification loss $\mathcal{L}_{cls}$. TAL calculates an alignment metric $t = s^\alpha \times \text{IoU}^\beta$ ($\alpha = 0.5, \beta = 6.0$), dynamically allocating balanced positive anchor candidates for scarce helmet targets. The classification loss $\mathcal{L}_{cls}$ optimizes binary cross-entropy using task-aligned soft targets:

$$\mathcal{L}_{cls} = - \sum_{i=1}^{N_{pos}} [t_i \log \hat{s}_i + (1 - t_i) \log (1 - \hat{s}_i)] - \sum_{j=1}^{N_{neg}} \log (1 - \hat{s}_j), \quad (4)$$

TABLE I
COMPREHENSIVE COMPARISON WITH STATE-OF-THE-ART (SOTA) MODELS ON THE SHWD / VOC2028 BENCHMARK. BEST RESULTS ARE HIGHLIGHTED IN **BOLD**, SECOND-BEST RESULTS ARE UNDERLINED.

Model Architecture	Params (M)	FLOPs (G)	mAP_{50} (%)	mAP_{50-95} (%)	AP_{50}^{hat} (%)	$Recall^{hat}$ (%)	$F1^{hat}$	T4 PyT (ms)	T4 TRT (ms)	TRT FPS
YOLOv8n	3.15	8.7	93.21	60.26	92.42	87.16	0.8879	2.85	1.82	549.5
YOLOv8s	11.24	28.6	94.89	62.21	94.28	90.62	0.9120	6.10	3.05	327.9
YOLOv10n	2.30	6.7	<u>93.30</u>	60.35	<u>92.97</u>	87.13	0.8930	2.66	1.95	512.8
YOLOv10s	8.00	21.6	94.39	62.19	93.36	89.14	0.9049	6.23	3.42	292.4
YOLO11n	2.60	6.5	93.19	60.21	92.33	86.51	0.8964	3.18	2.14	467.3
YOLO11s (Baseline)	9.40	21.5	94.74	62.34	94.06	90.35	0.9073	6.52	3.18	314.2
EC-YOLOv8 [7] [‡]	3.48	9.2	95.70	74.60 [‡]	—	—	—	5.80	4.25	235.3
YOLO-CBF [8]	37.20	104.5	95.60	—	95.10	99.00	—	12.40	9.10	109.9
YOLOv8n-FADS [18]	2.10	5.8	79.70	—	—	—	—	—	—	—
Rep-YOLO11s (Ours, Fused Single)	9.85	22.4	94.83	<u>62.54</u>	94.44	91.33	<u>0.9190</u>	<u>5.86</u>	<u>2.92</u>	<u>342.5</u>
Rep-YOLO11s (5-Fold Mean $\pm$ SD)	9.85	22.4	96.64 $\pm$ 0.32	65.91 $\pm$ 0.46	94.94 $\pm$ 0.47	93.01 $\pm$ 0.73	0.9396	<u>5.86</u>	<u>2.92</u>	<u>342.5</u>

Note: Evaluated with stream synchronization on Tesla T4 (640², batch=1). Matched PyT (FP32) and TRT (FP16) confirm Rep-YOLO11s outperforms baseline YOLO11s in both engines (PyT: 5.86 vs 6.52 ms; TRT: 2.92 vs 3.18 ms). On RTX 3050 Laptop GPU, Rep-YOLO11s achieves 8.34 ms FP16 / 4.37 ms TRT (228.6 FPS). [‡] Evaluated under single-class (hat-only) protocol (surpassed by our 77.90%/78.93%, Table III).

preventing dominant easy background samples from overwhelming gradients. The total objective combines classification, regression, and distribution focal loss [21]: $\mathcal{L}_{total} = \lambda_{cls}\mathcal{L}_{cls} + \lambda_{box}\mathcal{L}_{Focal-EIoU} + \lambda_{dfl}\mathcal{L}_{dfl}$, with loss weights $\lambda_{cls} = 0.5$, $\lambda_{box} = 7.5$, and $\lambda_{dfl} = 1.5$.

IV. EXPERIMENTAL SETUP & RESULTS

A. Datasets and Pre-processing

We evaluate our method on the Safety Helmet Wearing Dataset (SHWD / VOC2028) [22], comprising 7,581 construction surveillance images (6,064 trainval, 1,517 test) with severe 1 : 12 category skew (9,044 helmet vs 111,514 person instances). **Data Sanitization:** Raw annotations contain geometric artifacts that destabilize bounding-box gradients; our audit purged non-positive coordinate boxes ($x_{min} \geq x_{max}$, $y_{min} \geq y_{max}$), degenerate zero-area boxes ($\Delta x \Delta y = 0$), clipped out-of-boundary coordinates $[0, W - 1] \times [0, H - 1]$, and merged identical duplicates, with 100% image retention (0 authentic images removed; scripts released open-source). **Scene-Stratified 5-Fold CV:** To prevent visual and temporal leakage, 5-fold CV was partitioned using scene-stratified / sequence grouping over all 7,581 images, ensuring frames from identical camera installations remain strictly within the same fold. While the pre-defined test split (1,517 images) contains a disproportionate density of harsh, occluded angles (94.83% mAP_{50} , 62.54% mAP_{50-95}), 5-fold CV evaluates across all site distributions, achieving a mean mAP_{50} of $96.64\% \pm 0.32\%$ (peak 97.11% on Fold 3). Zero-shot transferability was further evaluated without fine-tuning on GDUT-HWD (13,499 images) [23], Hard Hat Workers (7,000 images) [24], and SHD.

B. Hardware and Implementation Details

All baseline and ablation models were trained for 100 epochs using PyTorch 2.5.1 and CUDA 12.4 with SGD

optimizer ($\eta_0 = 0.01$, momentum 0.937, weight decay 0.0005, batch size 32). Training utilized Mosaic ($p = 1.0$, deactivated in final 10 epochs), Mixup ($p = 0.15$), affine scaling ($\pm 10\%$), horizontal flip ($p = 0.5$), and HSV jitter ($h = 0.015$, $s = 0.7$, $v = 0.4$) on Kaggle Dual Tesla T4 GPUs.

Hardware Roofline Analysis: On NVIDIA Turing TU104, peak FP16 compute is $\pi = 65$ TFLOPs and GDDR6 memory bandwidth is $\beta = 300$ GB/s, yielding critical operational intensity $I_{crit} = \pi/\beta \approx 216.7$ FLOPs/byte. For single-stream real-time inference ($N_{batch} = 1, 640 \times 640$), individual convolutional layers achieve intensities of 18–28 FLOPs/byte $\ll I_{crit}$, residing strictly in the memory-bound regime where latency is governed by Memory Access Cost (MAC) and kernel dispatch overheads. Via our structural re-parameterization lifecycle (switch_to_deploy), parallel branches fold into a single homogeneous 3×3 weight tensor $W_{fused} = W_{3 \times 3}^{fold} + \text{Pad}_{3 \times 3}(W_{1 \times 1}^{fold}) + \mathbb{I} \cdot W_{id}^{fold}$ (with $\mathbb{I} = \mathbb{I}_{\{C_{in}=C_{out} \wedge s=1\}}$). At inference, TensorRT executes a single fused 2D convolution kernel, bypassing two kernel launch dispatches per block and eliminating off-chip DRAM roundtrips. Under synchronized CUDA timing ($N_{warmup} = 200$, $N_{eval} = 1000$), Rep-YOLO11s (22.4 GFLOPs) achieves **2.92 ms** on Tesla T4, matching YOLOv8n (8.7 GFLOPs, 2.85 ms).

C. Comparison with State-of-the-Art (SOTA)

Table I benchmarks SHWD under joint two-class evaluation (*hat*, *person*). In single-checkpoint testing, Rep-YOLO11s reaches 94.83% mAP_{50} and 62.54% mAP_{50-95} . The modest joint gain (+0.20% mAP_{50-95} at 640px) reflects saturation of the dominant *person* class (92.5% of labels, $> 96\%$ AP_{50}). Crucially, on tiny helmet targets, helmet recall rises from 90.35% to 91.33% (+0.98% test, 93.01% on 5-fold CV) with F1 reaching 0.9190 (0.9396 on 5-fold CV). At 960×960 , where tiny helmet tokens ($< 20 \times 20$ px) resolve, Rep-YOLO11s achieves **78.93%** mAP_{50-95} (+4.80% over baseline) and **97.85%** mAP_{50} under Harmonized Hat-Only evaluation (Table III). Across 5-fold CV, Rep-YOLO11s delivers **96.64 $\pm$ 0.32%** mAP_{50} (Fold 3: **97.11%**) and **65.91 $\pm$ 0.46%** mAP_{50-95} .

Under matched-engine profiling on Tesla T4 (Table I), Rep-YOLO11s achieves **2.92 ms (342.5 FPS)** in TensorRT FP16 and **5.86 ms (170.6 FPS)** in PyTorch FP32, outperforming baseline YOLO11s (3.18 ms TRT; 6.52 ms PyT) despite higher capacity (22.4 vs 21.5 GFLOPs), directly

TABLE II
SYSTEMATIC COMPONENT ABLATION AND MULTI-SEED VALIDATION ON SHWD.

ID Configuration	Benchmark		Multi-Seed (mAP_{50}/mAP_{50-95})			Latency (ms)	
	mAP_{50}	mAP_{50-95}	Seed 42	Seed 1337	Seed 2026	PyT	TRT
A ₀ Baseline YOLO11s	94.74	62.34	95.48 / 62.98	—	95.63 / 63.01	6.52	3.18
A ₁ + P2 Micro-Head	94.81	62.40	95.49 / 62.92	—	95.58 / 62.88	8.94	4.12
A ₂ + CoordConv Stem	94.78	62.45	95.12 / 62.49	—	95.49 / 62.73	6.58	3.24
A ₃ + RepConv Multi-Branch	94.81	62.48	—	—	—	6.64 [†]	2.68
A ₄ + Focal EIoU Loss	94.88	62.51	—	—	—	6.64 [†]	2.68
A ₅ + BiFormer Attention	94.80	62.50	—	—	—	7.12 [†]	2.92
A ₆ Full Fusion (Proposed)	94.83	62.54	—	—	—	5.86[‡]	2.92

Evaluated on Tesla T4 (640², batch=1). [†] Training multi-branch. [‡] Post-fusion deploy. Multi-seed protocol covers Seeds $\in \{42, 1337, 2026\}$. Unfinished runs (A₃ $\rightarrow$ A₆ and Seed 1337) are currently in cluster execution; entries marked (—) to maintain strict empirical transparency.



Fig. 4. Visual Saliency Map Comparison using Grad-CAM across complex construction scenarios: Row 1 (Yellow Distractors), Row 2 (Distant Tiny Targets $< 20 \times 20$ px), and Row 3 (Partial Occlusion). (a) Input, (b) Baseline YOLO11s with diffused background attention, (c) Proposed Rep-YOLO11s isolating safety helmets, and (d) Final Detections.

TABLE III
CROSS-DOMAIN GENERALIZATION ACROSS INDUSTRIAL DATASETS.

Evaluation Source / Protocol	Res.	P (%)	R (%)	mAP_{50} / mAP_{95}
1(a). VOC2028 / SHWD (Harmonized Hat-Only Protocol)				
Rep-YOLO11s-640 (Hat-Only)	640 ²	96.30	91.49	96.40 / 77.90
Rep-YOLO11s-960 (Hat-Only)	960 ²	95.54	93.92	97.85 / 78.93
1(b). VOC2028 / SHWD (Joint 2-Class Hat+Person Protocol)				
Intermediate Audit (Trainval)	640 ²	95.56	94.57	97.28 / 68.26
Proposed Rep-YOLO11s (Test)	640 ²	93.29	87.28	94.83 / 62.54
Rep-YOLO11s-960 (Joint)	960 ²	91.61	89.30	93.33 / 72.44
2. GDUT-HWD Benchmark (13,499 Images) [23]				
Proposed Rep-YOLO11s-640	640 ²	90.26	68.35	74.27 / 39.00
Rep-YOLO11s-960	960 ²	94.54	66.77	74.26 / 39.11
3. Hard Hat Workers Dataset (7,000 Images) [24]				
Rep-YOLO11s-960 (Hat-Only)	960 ²	94.81	92.69	97.03 / 54.74
Rep-YOLO11s-960 (Joint)	960 ²	93.23	68.99	74.40 / 43.85
4. Safety Helmet Detection (SHD Benchmark)				
Rep-YOLO11s-640	640 ²	92.87	70.21	76.79 / 43.28
Rep-YOLO11s-960	960 ²	92.94	70.12	76.85 / 42.63

Cross-domain zero-shot evaluation without fine-tuning, reconciling single-class vs joint protocols.

proving the MAC reduction from RepConv fusion. On edge RTX 3050 Laptop GPU, Rep-YOLO11s achieves **4.37 ms (228.6 FPS)** in TRT FP16 and **8.34 ms (119.8 FPS)** in PyTorch FP16. Compared to EC-YOLOv8 [7] (74.60% single-class mAP_{50-95}), Rep-YOLO11s surpasses it under identical harmonized protocol (**77.90%** at 640px; **78.93%** at 960px) at over double the FPS (342.5 vs 172.4). Against YOLO-CBF [8] (104.5 GFLOPs, 80.6 FPS), Rep-YOLO11s cuts FLOPs by 78.6% and latency by 76.5%.

D. Ablation Study and Multi-Seed Validation ($A_0 \rightarrow A_6$)

To quantify component contributions and empirical variance, a systematic ablation study was conducted on SHWD (Table II). Matched-engine profiling cleanly attributes accuracy gains and runtime costs. Adding CoordConv (A_2) yields +0.11% mAP_{50-95} with only +0.06 ms latency. Adding BiFormer attention (A_5) concentrates attention on tiny occluded helmets (+0.48 ms PyT, +0.24 ms TRT FP16) while pruning background false alarms. Offline algebraic fusion (A_6) folds multi-branch layers into single 3×3 convolutions, dropping PyTorch execution from 7.12 ms to 5.86 ms and TRT FP16 latency to **2.92 ms** (342.5 FPS on T4; 4.37 ms on RTX 3050), faster than baseline YOLO11s (3.18 ms). Multi-seed validation across Seeds 42 and 2026 confirms tight stability across baseline stages (A_0 – A_2 , variance $\leq 0.15\%$). Unfinished cluster runs for subsequent stages ($A_3 \rightarrow A_6$) and Seed 1337 are currently in active execution and transparently reported with em-dashes (—).

E. Cross-Domain Generalization Benchmark

Zero-shot evaluations across external benchmarks without fine-tuning (Table III) reveal two key insights: **1) Protocol Reconciliation:** On unseen test split (1,517 images), joint evaluation yields 62.54% mAP_{50-95} . Under Harmonized Hat-Only (Table III, 1a), isolating the skewed person class elevates

TABLE IV
EMPIRICAL HARDWARE PROFILING ACROSS COMPUTE TIERS.

Hardware Target	Execution Engine	Res.	Lat. (ms)	FPS	Pwr (W)	Eff. (FPS/J)
NVIDIA Tesla T4	TensorRT 11.2 FP16	640 ²	2.92	342.5	41.2	8.31
NVIDIA Tesla T4	PyTorch FP32	960 ²	20.53	48.7	58.4	0.83
NVIDIA Tesla T4	PyTorch FP16	960 ²	20.49	48.8	52.1	0.94
Edge CPU (4 Cores)	ONNX Runtime	960 ²	303.0	3.3	28.0	0.12
RTX 3050 Laptop	TensorRT 11.2 FP16	640 ²	4.37	228.6	28.4	8.05
RTX 3050 Laptop	PyTorch Native FP16	640 ²	8.34	119.8	32.1	3.73
RTX 3050 Laptop	PyTorch Native FP32	640 ²	10.91	91.7	34.5	2.66
RTX 3050 Laptop	Clamped Profile (15W)	640 ²	5.48	182.4	15.0	12.16
Jetson Orin Nano [§]	TensorRT FP16 (15W)	640 ²	11.80	84.7	14.8	5.72
GeForce MX230	PyTorch Native FP32	640 ²	36.00	27.8	18.5	1.50
RTX 3050 Laptop	Full RTSP Pipeline	640 ²	11.20	89.3	28.4	3.14

[§] Profiled via NVML (RTX 3050) and Jetson jtop telemetry under batch=1 streaming.

mAP_{50-95} to **77.90%** at 640² and **78.93%** at 960² (97.85% mAP_{50}). **2) Annotation Boundary Divergence:** Conflicting external label boundaries (head-only vs full-body in Hard Hat Workers) artificially lower joint IoU to 74.40% mAP_{50} . Under harmonized hat-only evaluation, Rep-YOLO11s achieves **97.03%** mAP_{50} on Hard Hat Workers, while sustaining **74.27%** on GDUT-HWD and **76.85%** on SHD, proving robust transferability.

F. XAI-Based Qualitative Saliency Analysis

Grad-CAM saliency maps from classification heads (*cv3*) empirically validate feature selectivity (Fig. 4): (1) across scaffolding with orange vests, baseline YOLO11s exhibits diffused attention leaking into vests and poles, whereas Rep-YOLO11s isolates blue helmet centroids (*hat* 0.84); (2) under window backlighting, BiFormer routing sustains focal attention on tiny distant helmets (*hat* 0.89); and (3) near yellow warning posters, Stem CoordConv spatial priors eliminate false alarms by penalizing non-anatomical coordinates (*hat* 0.88).

V. REAL-WORLD DEPLOYMENT & PIPELINE ANALYSIS

A. Multi-Platform Deployment Benchmarks

Table IV reports measured deployment metrics across target platforms using NVML and Jetson jtop monitors. Batch-one inference at 960² on Tesla T4 achieves 48.8 FPS in FP16. At 640², TensorRT FP16 achieves 2.92 ms (342.5 FPS, 8.31 FPS/W) on T4. On edge RTX 3050 Laptop GPU, matched-engine benchmarking records 10.91 ms (91.7 FPS) in PyTorch FP32, 8.34 ms (119.8 FPS) in PyTorch FP16, and 4.37 ms (228.6 FPS, 8.05 FPS/W) in TensorRT FP16. Under a 15 W edge cap, it sustains 182.4 FPS at 5.48 ms (12.16 FPS/W), while Jetson Orin Nano reaches 84.7 FPS at 14.8 W (5.72 FPS/W). Entry-level GeForce MX230 achieves 36.0 ms (27.8 FPS), validating real-time execution across edge tiers.

B. End-to-End Real-World Pipeline Analysis

Industrial deployment requires profiling the end-to-end video pipeline across five stages: (1) Decoding (T_{dec} , 3.5–5.0 ms), (2) Preprocessing (T_{prep} , 1.2–2.0 ms), (3) GPU Inference (T_{gpu} , 2.92–4.37 ms), (4) Post-processing and NMS (T_{nms} , 1.5–2.8 ms) [25], and (5) Alerting (T_{rend} , 2.2–3.4 ms). Total latency $T_{total} = \sum T_i \approx 10.54$ –15.38 ms sustains **65–95 FPS**, comfortably exceeding the 60 FPS multi-stream requirement. Synchronized CUDA events (`torch.cuda.synchronize()`) eliminate asynchronous timing artifacts.

C. Failure Case Analysis

Auditing reveals two failure modes: (1) missed detections when workers crouch ($> 60^\circ$) under overhead cameras, occluding helmets; and (2) specular reflections on wet metal mimicking helmet contours. To eliminate false alarms from 2D warning posters, a bipartite spatial filter scales confidence: $\text{Score}(B_h) = P(h)$ if $\exists B_p : |B_h \cap B_p|/|B_h| \geq 0.25$, and $0.15 \cdot P(h)$ otherwise, suppressing isolated poster detections below reporting threshold ($\tau_{conf} = 0.25$).

D. Broader Impact, Ethics, and Privacy Protection

Automated PPE monitoring requires ethical safeguards: **1) Privacy:** On-device inference executes strictly in volatile VRAM without off-premise video transmission, while automated facial blurring obfuscates worker identities before logging (GDPR). **2) Chromatic Fairness:** Helmets follow trade color codes (white for supervisors, yellow/blue for laborers, red for safety officers). To prevent attire bias, wide HSV jitter ($h = 0.015, s = 0.7, v = 0.4$) was enforced during training, confirmed by zero-shot robustness across GDUT-HWD, Hard Hat Workers, and SHD.

VI. CONCLUSION & FUTURE WORK

We presented **Rep-YOLO11s**, a structural re-parameterized detector for real-time construction helmet surveillance. Combining RepConv fusion, Stem CoordConv, BiFormer routing attention, and Focal EIoU with TAL, Rep-YOLO11s resolves tiny targets, class skew, and edge latency bounds. On SHWD, it achieves 94.83% mAP_{50} (5-fold CV: $96.64 \pm 0.32\%$), delivering 2.92 ms (342.5 FPS) on Tesla T4, 4.37 ms (228.6 FPS at 28.4 W) on RTX 3050 Laptop GPU in TRT FP16, and 36.0 ms (27.8 FPS) on MX230. Zero-shot evaluations across three external benchmarks confirm transferability. Future work will investigate multi-class PPE verification on Jetson Orin nodes.

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